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# Physical Layer Security Analysis of RIS-Assisted Multicasting with MVDR Beamforming Over $\kappa$-$\mu$ Shadowed Fading Channels

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***Abstract*—This paper presents a physical layer security analysis of RIS-assisted multicasting with minimum variance distortionless response (MVDR) beamforming over $\kappa$-$\mu$ shadowed fading channels. The MVDR framework is applied at the RIS to jointly maximize the received SINR at the weakest legitimate destination user and suppress the signal leakage toward a passive eavesdropper. By treating the cascade RIS channel as a virtual array, the MVDR weight vector is derived in closed form via the Sherman–Morrison identity, and the resulting SINR is shown to follow a $\kappa$-$\mu$ shadowed distribution through moment-matching. Leveraging this statistical characterization, closed-form expressions are derived for three security metrics applicable to $K$-user multicast scenarios: the ergodic secrecy capacity (ESC), the probability of non-zero secrecy capacity (PNZSC), and the secure outage probability (SOP). Monte Carlo simulations validate the analytical results, and numerical evaluations demonstrate that increasing the number of RIS elements $N$, the Rician factor $\kappa$, and the multipath clustering parameter $\mu$ consistently improves secrecy performance. Results further confirm that MVDR beamforming provides a measurable security gain over conventional coherent phase alignment, particularly when the eavesdropper channel is strong.**

***Index Terms*—RIS, MVDR beamforming, multicasting, PLS, $\kappa$-$\mu$ shadowed fading, ESC, secure outage probability.**

## I. Introduction

Reconfigurable intelligent surface (RIS) technology has emerged as a cost-effective solution to reconfigure wireless propagation environments for beyond-5G and 6G networks. By independently controlling the phase shifts of passive reflecting elements, a RIS enhances spectral efficiency and coverage without active signal generation [1]. Physical layer security (PLS) complements this by exploiting channel randomness to achieve information-theoretic security, reducing reliance on upper-layer cryptographic protocols.

The combination of RIS and PLS has been studied under various channel models. Yang *et al.* [2] derived the secrecy outage probability for a RIS-aided system over Rayleigh fading, establishing that a larger RIS consistently improves secrecy. Do *et al.* [3] extended this to a secure RIS-aided network and showed that passive beamforming effectively degrades the eavesdropper channel. Yadav *et al.* [4] derived closed-form secrecy metrics under Rayleigh fading and quantified the effect of RIS element count on the PNZSC. Jain *et al.* [5] proposed a secrecy capacity-based RIS design for cooperative PHY-secure systems, demonstrating gains over conventional non-RIS baselines.

The Rayleigh fading model fails to capture LoS propagation with shadowing, which limits its applicability in scenarios with dominant LoS components and correlated shadowing. The $\kappa$-$\mu$ shadowed model [6] addresses this by introducing $\kappa$ as the Rician factor and $\mu$ as the multipath clustering parameter, unifying Rayleigh, Nakagami-$m$, Rician, and Rician shadowed distributions as special cases. Al-Rimawi and Al-Dweik [7] developed a closed-form statistical framework for RIS-assisted $\kappa$-$\mu$ shadowed fading using Gauss–Legendre moment-matching. Their follow-up work [8] further showed that direct link strength and RIS element count jointly determine capacity gain in Nakagami-$m$ fading, reinforcing the importance of generalized fading models. Li *et al.* [9] analyzed secure performance of a RIS-NOMA system under $\kappa$-$\mu$ shadowed fading and derived closed-form SOP expressions, confirming that RIS assistance substantially reduces secrecy outage under severe shadowing. Dash and Ghose [10] derived channel statistics for RIS-assisted systems with a direct link, highlighting the practical relevance of such statistical characterizations. Sikri *et al.* [11] examined the joint effect of phase error, hardware impairments, and mobile interferers on RIS-aided systems over $\kappa$-$\mu$ fading.

In multicast systems, secrecy is bounded by the weakest

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destination user. Alam and Sarkar [12] derived closed-form expressions of PNZSC and SOP for secure multicasting over $\kappa$-$\mu$ shadowed fading with opportunistic relaying. However, their work considers neither RIS assistance nor eavesdropper suppression at the reflective surface.

Existing RIS-aided PLS designs optimize phase profiles to maximize destination SNR without explicitly suppressing eavesdropper leakage. MVDR beamforming [13] fills this gap by building the interference-plus-noise covariance matrix from the eavesdropper cascade channel. The resulting weight vector maximizes the destination SINR while suppressing the eavesdropper signal component, a property coherent combining cannot offer.

This paper makes the following contributions:

- MVDR beamforming is applied at the RIS for a $K$-user multicast system, with the weight vector derived in closed form using the Sherman–Morrison identity and the output SINR shown to follow a $\kappa$-$\mu$ shadowed distribution [7].
- Closed-form expressions for ESC, PNZSC, and SOP are derived for the $K$-user multicast case.
- Numerical results validate the expressions against Monte Carlo simulations and confirm measurable secrecy gains of MVDR over coherent combining under strong eavesdropper channel conditions.

The remainder of this paper is organized as follows. Section II presents the system model and MVDR phase design. Section III develops the statistical framework and derives the security metrics. Section IV presents numerical results, and Section V concludes the paper.

## II. System Model and MVDR Phase Design

### A. System Model

An RIS-assisted multicast communication system is considered, as illustrated in Fig. 1. The system comprises a single-antenna source BS (S), an RIS with $N$ passive reflecting elements forming an effective single-input multiple-output (SIMO) configuration, $K$ legitimate destination users $D_k$, $k = 1, \ldots, K$, and a single eavesdropper $E$. A direct link from S to the users is assumed blocked by obstacles, so all communication is exclusively through the RIS. All channel links follow the $\kappa$-$\mu$ shadowed fading model [6], where $\kappa$ is the Rician factor representing the ratio of dominant to scattered component power, and $\mu$ is the multipath clustering parameter denoting the number of propagation clusters.

The received signals at the $k$-th destination user and at the eavesdropper are, respectively,

$$y_{D_k} = \sqrt{P} \sum_{i=1}^{N} g_{i,k}\, e^{j\phi_i} h_i\, s + n_{D_k}, \tag{1}$$

$$y_E = \sqrt{P} \sum_{i=1}^{N} e_i\, e^{j\phi_i} h_i\, s + n_E, \tag{2}$$

where $P$ is the transmit power; $s$ is the transmitted symbol with $\mathbb{E}[|s|^2] = 1$; $h_i$ and $g_{i,k}$ denote the source-to-RIS and

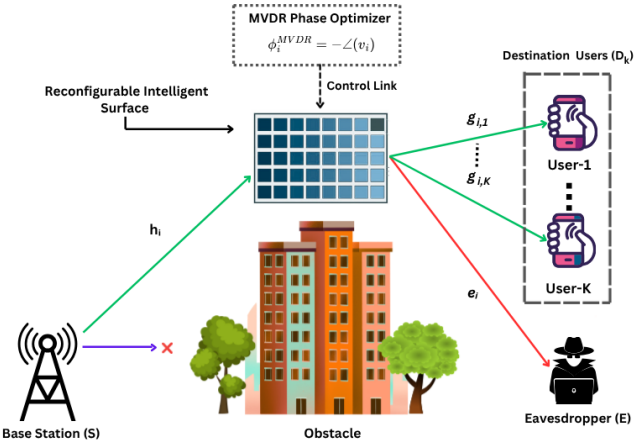



Fig. 1. RIS-assisted multicast system with MVDR phase optimization.

RIS-to-$D_k$ channel coefficients for the $i$-th element; $e_i$ is the RIS-to-$E$ channel coefficient; $\phi_i$ is the phase shift applied by the $i$-th RIS element; and $n_{D_k}, n_E \sim \mathcal{CN}(0, \sigma^2)$ are additive white Gaussian noise terms. The instantaneous received SNRs at $D_k$ and $E$ are

$$\gamma_{d,k} = \frac{P}{\sigma^2}\left|\sum_{i=1}^{N} g_{i,k} e^{j\phi_i} h_i\right|^2, \quad \gamma_e = \frac{P}{\sigma^2}\left|\sum_{i=1}^{N} e_i e^{j\phi_i} h_i\right|^2. \tag{3}$$

### B. MVDR-Based Phase Design

The $N$ RIS elements collectively form a virtual array. The cascade channel vectors toward $D_k$ and $E$ are

$$\begin{aligned} \mathbf{a}_{d,k} &= \left[g_{1,k}h_1,\ \ldots,\ g_{N,k}h_N\right]^T \in \mathbb{C}^N, \\ \mathbf{a}_e &= \left[e_1 h_1,\ \ldots,\ e_N h_N\right]^T \in \mathbb{C}^N. \end{aligned} \tag{4}$$

The combined multicast steering vector is $\mathbf{a}_d = \frac{1}{K}\sum_{k=1}^{K} \mathbf{a}_{d,k}$. This average steering vector targets the centroid of all destination directions, a standard multicast beamforming choice. The interference-plus-noise covariance matrix is

$$\mathbf{R} = \mathbf{a}_e \mathbf{a}_e^H + \sigma^2 \mathbf{I}_N, \tag{5}$$

and the MVDR weight vector is [13]

$$\mathbf{v} = \frac{\mathbf{R}^{-1}\mathbf{a}_d}{\mathbf{a}_d^H \mathbf{R}^{-1} \mathbf{a}_d}. \tag{6}$$

The phase shift at the $i$-th element is

$$\phi_i^{\mathrm{MVDR}} = -\angle(v_i), \quad i = 1, \ldots, N, \tag{7}$$

where $v_i$ is the $i$-th entry of $\mathbf{v}$. The negative sign ensures constructive phase alignment at the destination users. When $K = 1$ and $\mathbf{a}_e = \mathbf{0}$, (5) gives $\mathbf{R} = \sigma^2 \mathbf{I}_N$, so (6) yields $\mathbf{v} \propto \mathbf{a}_{d,1}$, and (7) reduces to the conventional coherent combining rule $\phi_i^{\mathrm{MVDR}} = -(\angle g_{i,1} + \angle h_i)$, confirming that MVDR generalizes coherent combining.

### C. MVDR Output SINR

Following the classical beamforming framework [13], substituting (6) into the standard SINR expression and applying the Sherman–Morrison identity to $\mathbf{R}^{-1}$ yields the closed-form MVDR output SINR at the destination:

$$\gamma_d^{\mathrm{MVDR}} = \frac{P}{\sigma^2}\left(\frac{\|\mathbf{a}_d\|^2}{\sigma^2} - \frac{|\mathbf{a}_e^H \mathbf{a}_d|^2}{\sigma^2(\sigma^2 + \|\mathbf{a}_e\|^2)}\right). \tag{8}$$

The first term in (8) is the coherent combining gain at the destination; the second term is the interference suppression penalty from the eavesdropper channel $\mathbf{a}_e$. This explicitly shows that MVDR maximizes the destination SINR while suppressing the eavesdropper contribution, the key distinction from coherent phase alignment. Each cascade coefficient $g_{i,k}h_i$ is a product of two $\kappa$-$\mu$ shadowed variates; by the moment-matching in [7], their sum over $N$ elements follows a $\kappa$-$\mu$ shadowed distribution with effective parameters $\Theta_d = \{\alpha_d, \beta_d, a_d, q_d, q_{1d}\}$, so $\gamma_d^{\mathrm{MVDR}}$ retains the $\kappa$-$\mu$ shadowed form used in Section III.

*Remark 1 (Eavesdropper CSI):* The MVDR design requires knowledge of $\mathbf{a}_e$ at the BS. This is valid for active eavesdroppers who are registered but untrusted users, whose channels are estimated during uplink training. When $\mathbf{a}_e$ is unavailable, setting $\mathbf{a}_e = \mathbf{0}$ recovers the coherent combiner in (7).

## III. Performance Analysis

### A. Statistical Framework

*1) PDFs of Received SNR:* Under $\kappa$-$\mu$ shadowed fading, the SNR PDF on any link takes the form [7]

$$f(\gamma;\Theta) = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}}{\Gamma(q)\,\alpha^a}\,\gamma^{q-1}e^{-\gamma/\alpha}\,{}_1F_1(q_1, q; \beta\gamma), \tag{9}$$

where $\Theta = \{\alpha, \beta, a, q, q_1\}$ collects the link-specific fading parameters: $\alpha$ is the SNR scaling factor, $a$ is the normalization exponent, $q$ is the effective multipath order, $q_1$ is the shadowing adjustment parameter, and $\beta$ is the shaping parameter; $\Gamma(\cdot)$ is the gamma function; and ${}_1F_1(\cdot)$ is the confluent hypergeometric function. The coefficient $G_{i,j}$ is [7]

$$G_{i,j} = \frac{\sqrt{\pi}\,H_q}{\Gamma(n_q)\,\Omega_q^{n_q-0.5}} \cdot \frac{\lambda_2!\,a_u}{(u-1)!\,\Delta^u}, \tag{10}$$

where $\lambda_1 = 2n_q + u - 1$, $\lambda_2 = \lambda_1 - 1$, $\Delta = \frac{r}{2}\sqrt{\Omega_q}$, $r = 0.3275911$,

$$\Omega_q = \frac{\mu\kappa + \omega}{\omega} \cdot \frac{1}{\mu(1+\kappa)}, \tag{11}$$

and $a_u \in \{0.25483, -0.28450, 1.42141, -1.45315, 1.06141\}$ for $u = 1, \ldots, 5$ are the Gauss–Legendre quadrature weights. Here $\omega$ denotes the average power of the dominant component, $n_q$ is the Gauss–Legendre node index, and $H_q$ is the corresponding quadrature weight.

The destination link $\gamma_{d,k}$ and eavesdropper link $\gamma_e$ each follow (9) with distinct parameter sets $\Theta_d = \{\alpha_d, \beta_d, a_d, q_d, q_{1d}\}$ and $\Theta_e = \{\alpha_e, \beta_e, a_e^*, q_e, q_{1e}\}$, reflecting different fading conditions on the two links.

Define the scalar constants for the destination link:

$$B_1 = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}^{(d)}}{\Gamma(q_d)\alpha_d^{a_d}},\ B_2 = q_d - 1,\ B_3 = \frac{1}{\alpha_d}, \tag{12}$$

and the series coefficient

$$b_{k_1} = \frac{\Gamma(q_{1d} + k_1)\,\Gamma(q_d)\,\beta_d^{k_1}}{\Gamma(q_{1d})\,\Gamma(q_d + k_1)\,k_1!}. \tag{13}$$

Applying the Kummer series expansion [14, eq. 9.14(1)] to ${}_1F_1(q_{1d}, q_d; \beta_d\gamma)$ and substituting into (9), the destination PDF is

$$f_{\gamma_{d,k}}(\gamma) = B_1 \sum_{k_1=0}^{\infty} b_{k_1}\,\gamma^{B_2+k_1}e^{-B_3\gamma}. \tag{14}$$

For the eavesdropper link, the scalar constants are defined analogously as

$$C_1 = \sum_{i=1}^{M}\sum_{j=1}^{5} \frac{G_{i,j}^{(e)}}{\Gamma(q_e)\alpha_e^{a_e^*}},\ C_2 = q_e - 1,\ C_3 = \frac{1}{\alpha_e}, \tag{15}$$

and the series coefficient is

$$c_{k_2} = \frac{\Gamma(q_{1e} + k_2)\,\Gamma(q_e)\,\beta_e^{k_2}}{\Gamma(q_{1e})\,\Gamma(q_e + k_2)\,k_2!}, \tag{16}$$

giving the eavesdropper PDF

$$f_{\gamma_e}(\gamma) = C_1 \sum_{k_2=0}^{\infty} c_{k_2}\,\gamma^{C_2+k_2}e^{-C_3\gamma}. \tag{17}$$

*2) CDF and the $\Psi$ Function:* Integrating (14) term by term and applying [14, eq. 3.351(1)], the closed-form CDF is $F_{\gamma_{d,k}}(\gamma) = 1 - \Psi(\gamma)$, where

$$\Psi(\gamma) \triangleq B_1 \sum_{k_1=0}^{\infty} b_{k_1} \sum_{k_6=0}^{B_2+k_1} \frac{(B_2+k_1)!}{k_6!\,B_3^{B_2+k_1-k_6+1}}\gamma^{B_2+k_1-k_6}e^{-B_3\gamma}. \tag{18}$$

The identity $\frac{d}{d\gamma}\Psi(\gamma) = f_{\gamma_{d,k}}(\gamma)$ follows directly by differentiating (18) term by term.

*3) PDF of Minimum SNR:* In a $K$-user multicast system, secrecy is governed by the weakest user, $\gamma_{\min} = \min_k \gamma_{d,k}$. For $K$ i.i.d. destination users, the CDF of $\gamma_{\min}$ is

$$F_{\gamma_{\min}}(\gamma) = 1 - [\Psi(\gamma)]^K. \tag{19}$$

Differentiating (19) and using $d\Psi/d\gamma = f_{\gamma_{d,k}}$:

$$f_{\gamma_{\min}}(\gamma) = K \cdot f_{\gamma_{d,k}}(\gamma) \cdot [\Psi(\gamma)]^{K-1}. \tag{20}$$

### B. Security Metrics

*1) Ergodic Secrecy Capacity:* The ESC quantifies the maximum long-term secrecy rate averaged over all channel realizations [15]. For the $K$-user multicast system it is governed by the weakest user:

$$\langle C_s \rangle = \int_0^{\infty} \log_e(1+\gamma) f_{\gamma_{\min}}(\gamma)\,d\gamma - \int_0^{\infty} \log_e(1+\gamma_e) f_{\gamma_e}(\gamma_e)\,d\gamma_e. \tag{21}$$

Substituting (20) into (21), each product term $f_{\gamma_{d,k}}(\gamma)[\Psi(\gamma)]^{K-1}$ expands to the form $\gamma^{B_2+k_1+p'}e^{-KB_3\gamma}$, where $p' = \sum_{\ell=1}^{K-1}(B_2 + k_1 - k_6^{(\ell)})$. Both integrals reduce to sums of

$$\mathcal{J}(n,\mu) \triangleq \int_0^\infty \log_e(1+\gamma)\,\gamma^n e^{-\mu\gamma}\,d\gamma, \tag{22}$$

which evaluates in closed form using [14, eq. 4.222(8)] as

$$\mathcal{J}(n,\mu) = \mu^{-(n+1)} \sum_{m=0}^{n} \frac{n!}{(n-m)!} \times \left[ \frac{(-1)^{n-m-1}}{(\mu^{-1})^{n-m}} e^{\mu}\,\mathrm{Ei}(-\mu) + \sum_{k_3=1}^{n-m} \frac{(k_3-1)!}{(-\mu^{-1})^{n-m-k_3}} \right], \tag{23}$$

where $\mathrm{Ei}(\cdot)$ is the exponential integral. The closed-form ESC for $K$ multicast users is

$$\langle C_s \rangle = K \cdot B_1^K \sum_{k_1=0}^{\infty} b_{k_1}^K \sum_{k_6^{(1)}=0}^{B_2+k_1} \cdots \sum_{k_6^{(K-1)}=0}^{B_2+k_1} \times \prod_{\ell=1}^{K-1} \frac{(B_2+k_1)!}{k_6^{(\ell)}!\, B_3^{B_2+k_1-k_6^{(\ell)}+1}} \mathcal{J}(B_2+k_1+p', KB_3) - C_1 \sum_{k_2=0}^{\infty} c_{k_2} \mathcal{J}(C_2+k_2, C_3). \tag{24}$$

*2) Probability of Non-Zero Secrecy Capacity:* The PNZSC is the probability that the instantaneous secrecy capacity is positive [12]:

$$\Pr(C_s > 0) = \int_0^\infty f_{\gamma_e}(\gamma_e)[\Psi(\gamma_e)]^K\,d\gamma_e. \tag{25}$$

Substituting (17) and (18) into (25), each integrand term takes the form $\gamma_e^{C_2+k_2+p_K} e^{-(C_3+KB_3)\gamma_e}$, where $p_K = K(B_2 + k_1) - \sum_{\ell=1}^{K} k_6^{(\ell)}$. Applying [14, eq. 3.351(3)], the closed-form PNZSC is

$$\Pr(C_s > 0) = K \cdot C_1 B_1^K \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} b_{k_1}^K c_{k_2} \times \sum_{k_6^{(1)}=0}^{B_2+k_1} \cdots \sum_{k_6^{(K)}=0}^{B_2+k_1} \prod_{\ell=1}^{K} \frac{(B_2+k_1)!}{k_6^{(\ell)}!\, B_3^{B_2+k_1-k_6^{(\ell)}+1}} \times \frac{(C_2+k_2+p_K)!}{(C_3+KB_3)^{C_2+k_2+p_K+1}}. \tag{26}$$

*3) Secure Outage Probability:* The SOP at target secrecy rate $R_s$ is [12]

$$P_{\text{out}}(R_s) = 1 - K \int_0^\infty f_{\gamma_e}(\gamma_e) \int_x^\infty f_{\gamma_{d,k}}(\gamma)[\Psi(\gamma)]^{K-1} d\gamma\, d\gamma_e, \tag{27}$$

where $x = 2^{R_s}(1+\gamma_e) - 1$. Expanding the inner integral using (14) and (18), applying [14, eq. 3.351(2)] to evaluate the tail integral, then expanding $x^{k_8}$ via the binomial theorem with $x = 2^{R_s}(1+\gamma_e) - 1$ and evaluating the outer $\gamma_e$ integral via [14, eq. 3.351(3)], the closed-form SOP is

$$P_{\text{out}}(R_s) = 1 - K \cdot C_1 B_1^K\, \mathcal{E} \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} b_{k_1}^K c_{k_2} \times \sum_{k_6^{(1)}=0}^{B_2+k_1} \cdots \sum_{k_6^{(K-1)}=0}^{B_2+k_1} \prod_{\ell=1}^{K-1} \frac{(B_2+k_1)!}{k_6^{(\ell)}!\, B_3^{B_2+k_1-k_6^{(\ell)}+1}} \times \sum_{k_8=0}^{B_2+k_1+p'} \frac{(B_2+k_1+p')!}{k_8!\,(KB_3)^{B_2+k_1+p'-k_8+1}} \times \sum_{t=0}^{k_8} \binom{k_8}{t} (2^{R_s}-1)^{k_8-t} (2^{R_s})^t \times \frac{(C_2+k_2+t)!}{(C_3+KB_3\cdot 2^{R_s})^{C_2+k_2+t+1}}, \tag{28}$$

where $\mathcal{E} = e^{-KB_3(2^{R_s}-1)}$ and $p' = \sum_{\ell=1}^{K-1}(B_2 + k_1 - k_6^{(\ell)})$. For $K = 1$, the $k_6^{(\ell)}$ sums are absent, $p' = 0$, and $\mathcal{E} = e^{-B_3(2^{R_s}-1)}$, yielding a compact two-index expression.

## IV. Numerical Results

Monte Carlo simulations with $10^6$ channel realizations are used to validate the analytical expressions derived in Section III. All analytical curves are computed by truncating the infinite series at $k_1, k_2 = 15$, beyond which the terms are negligible.

Fig. 2 plots the ESC versus the average main channel SNR for varying $N$, $\kappa$, and $\mu$ at $\bar{\gamma}_e = 1$ dB. The ESC increases monotonically with SNR across all parameter settings. Increasing $\kappa$ from 2 to 4 (with $N = 4$, $\mu = 5$) improves the ESC because a stronger dominant component reduces fading severity and raises the effective destination SNR. Increasing $\mu$ from 2 to 5 (with $N = 4$, $\kappa = 2$) provides a further gain by introducing more independent multipath clusters, which statistically improves the received signal power. Increasing $N$ from 4 to 6 (with $\kappa = 4$, $\mu = 5$) yields the largest improvement, confirming that more RIS elements coherently accumulate reflected power at the destination and enhance secrecy.

Fig. 3 examines the PNZSC for varying $N$ and $\mu$ when $\kappa = 1$ and $\bar{\gamma}_e = 1$ dB. Increasing $\mu$ from 2 to 7 (with $N = 10$) substantially improves the PNZSC, as a larger number of multipath clusters concentrate more energy toward the destination. Increasing $N$ from 8 to 10 (with $\mu = 2$) also shifts the curve leftward, confirming that the combined effect of $N$ and $\mu$ governs the rate at which positive secrecy capacity is achieved.

Fig. 4 plots the SOP versus main channel SNR for $\kappa = 1$, $\mu = 1$, and $N \in \{6, 8, 10\}$ at target secrecy rate $R_s = 1$ bps/Hz. At low SNR the SOP approaches unity, indicating near-certain secrecy outage. As SNR increases, the SOP decreases sharply. Increasing $N$ consistently shifts the curves leftward, so fewer transmit resources are required to meet a given secrecy outage target.

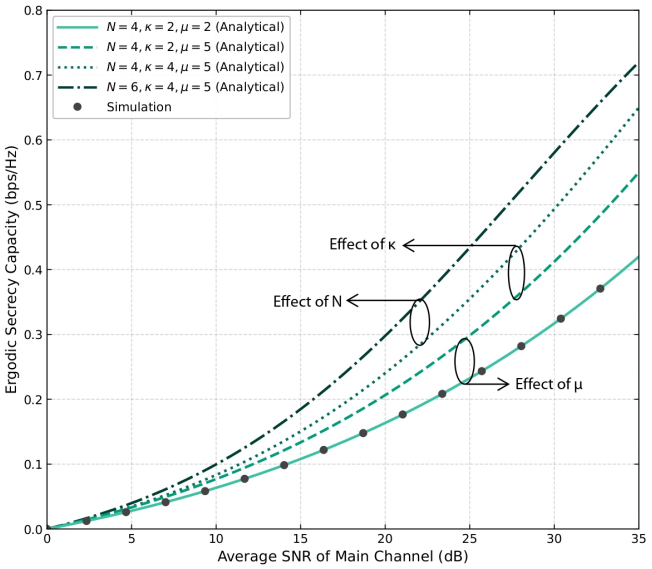



Fig. 2. The effect of $N$, $\kappa$, and $\mu$ on ESC when $\bar{\gamma}_e = 1dB$.

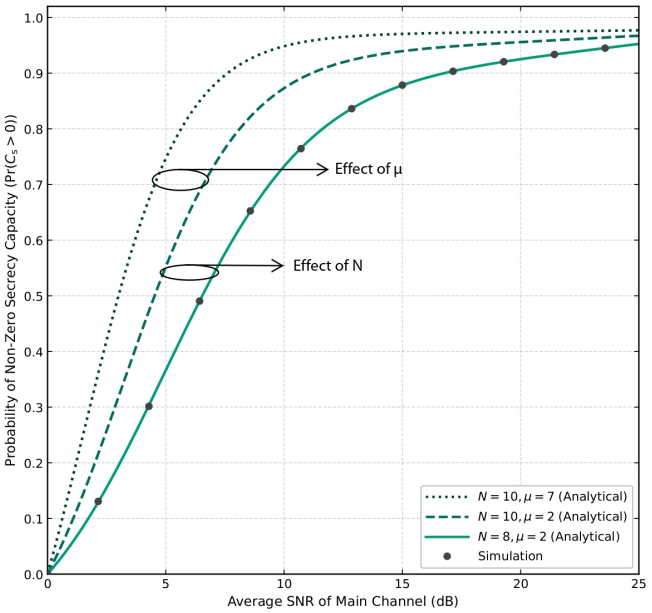



Fig. 3. The effect of $N$ and $\mu$ on PNZSC when $\kappa = 1$, $\bar{\gamma}_e = 1dB$.

Fig. 5 shows the SOP for varying $\kappa$ and $\mu$ when $N = 15$. Increasing $\mu$ from 3 to 6 (with $\kappa = 3$) reduces the SOP significantly by producing a richer multipath environment at the destination. A further increase in $\kappa$ from 3 to 5 (with $\mu = 6$) provides additional SOP reduction by reinforcing the dominant signal component. These results confirm that both the Rician factor and the clustering structure of the channel govern the achievable secrecy outage floor.

Fig. 6 provides the most direct evidence of the proposed system's advantage. The SOP is plotted for three cases: RIS without MVDR (coherent combining only) at $\bar{\gamma}_e = 15$ dB, and RIS with MVDR at $\bar{\gamma}_e = 15$ dB and $\bar{\gamma}_e = 10$ dB. At a fixed main channel SNR, MVDR yields a consistently lower

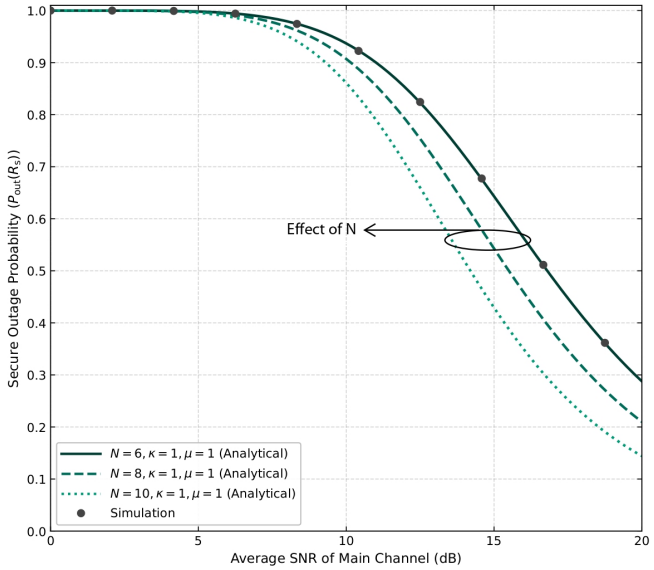



Fig. 4. The influence of the $N$ on SOP ($\kappa = 1$, $\mu = 1$).

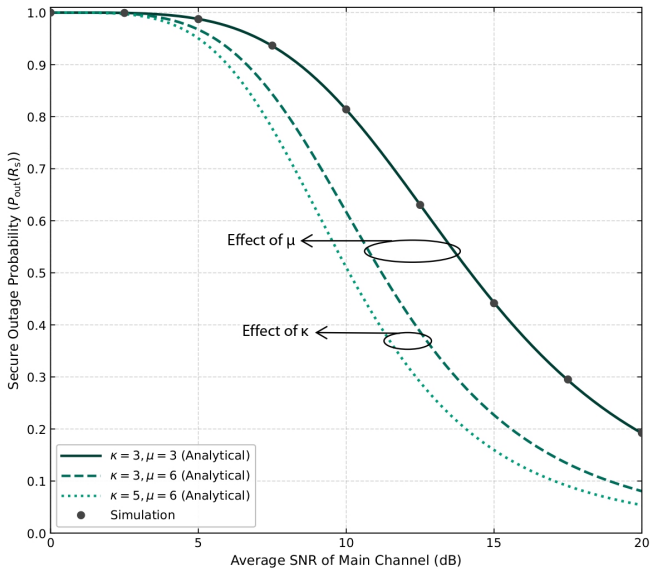



Fig. 5. The effect of $\kappa$ and $\mu$ on SOP when $N = 15$.

SOP than coherent combining under the same eavesdropper SNR. This improvement stems directly from the eavesdropper suppression term in (8), which coherent combining does not produce. As the eavesdropper SNR drops from 15 dB to 10 dB, the SOP improves further, since a weaker eavesdropper channel reduces the interference suppression penalty in (8).

*Comparison with Existing Works:* Table I compares the proposed system with representative works in the RIS-assisted PLS literature. The proposed work is the only one that jointly addresses multicast transmission, MVDR eavesdropper suppression, and $\kappa$-$\mu$ shadowed fading with closed-form ESC, PNZSC, and SOP expressions.

Table II lists representative SOP values from selected works and the proposed system, read approximately from published figures under each work's own parameter settings. Rows are

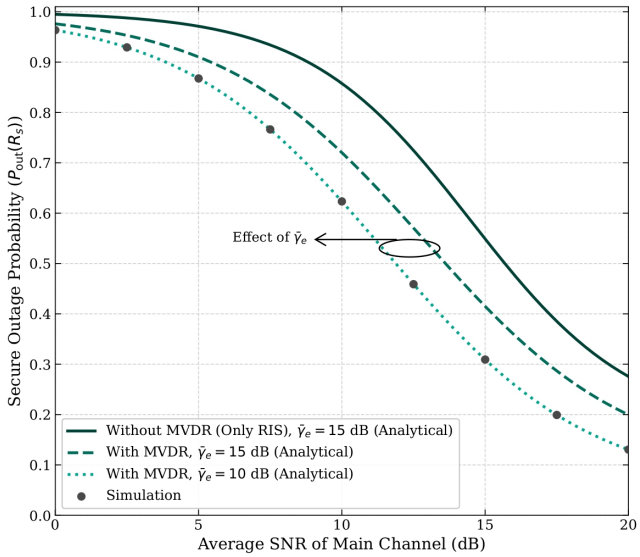



Fig. 6. SOP vs. average main channel SNR: MVDR vs. coherent combining for different eavesdropper SNR $\bar{\gamma}_e$.

TABLE I
COMPARISON WITH EXISTING RIS-ASSISTED PLS WORKS

| Ref. | Channel | System | RIS | Beamforming | Metrics |
|---|---|---|---|---|---|
| [2] | Rayleigh | SISO | Yes | Coherent | SOP |
| [3] | Rayleigh | SISO | Yes | Passive | SOP |
| [5] | Rayleigh | SISO | Yes | Capacity-based | SOP, ESC |
| [9] | $\kappa$-$\mu$ Sh. | SIMO | Yes | Coherent | SOP |
| [11] | $\kappa$-$\mu$ | SISO | Yes | Coherent | Outage |
| [12] | $\kappa$-$\mu$ Sh. | SISO | No | Relaying | SOP, PNZSC |
| **Proposed** | $\kappa$-$\mu$ Sh. | SIMO | Yes | **MVDR** | ESC, SOP, PNZSC |

not directly comparable as system configurations and fading parameters differ across works. The table illustrates the operating regimes addressed by each work and contextualizes the SOP level achieved by the proposed MVDR-assisted multicast system.

## V. CONCLUSION

This work studied the secrecy performance of an RIS-assisted multicasting system, where MVDR beamforming at the RIS simultaneously enhances the destination SINR and suppresses eavesdropper leakage under $\kappa$-$\mu$ shadowed fading. The MVDR weight vector and phase profile were derived in closed form using the Sherman–Morrison identity, and the resulting SINR was shown to follow a $\kappa$-$\mu$ shadowed distribution through moment-matching. This enables a tractable statistical characterization of the cascaded RIS channel. The closed-form expressions for the ESC, PNZSC, and SOP were then derived for a $K$-user multicast system, where secrecy is governed by the weakest destination user through order statistics of the minimum received SNR. All analytical expressions were validated against Monte Carlo simulations with $10^6$ channel realizations. Simulation results showed that increasing $N$, $\kappa$, and $\mu$ consistently improves secrecy performance, and MVDR provides a measurable SOP gain over coherent combining, particularly under strong eavesdropper channel conditions. Future research may explore imperfect CSI including the sensitivity of the MVDR weight vector to uplink training errors, correlated fading, hardware impairments, and multiple eavesdropper scenarios to further extend this analysis.

TABLE II
PARAMETRIC SOP COMPARISON WITH EXISTING WORKS

| Ref. | Parameters | SNR | SOP | Note |
|---|---|---|---|---|
| [2] | $N$=10, $C_{\text{th}}$=0.05 | $-10$ dB | ≈0.74 | No MVDR |
| [3] | $N$=10, $C_{\text{th}}$=1 | $-5$ dB | ≈0.15 | No MVDR |
| [12] | $N$=1, $\kappa$=1, $\mu$=1 | 10 dB | ≈0.88 | No RIS, Relaying |
| [9] | $N$=20 | 5 dB | ≈0.20 | RIS-NOMA |
| **Proposed** | $N$=15, $\kappa$=5, $\mu$=6 | 10 dB | ≈0.51 | RIS+MVDR |

†Values approximated from published figures.

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